SEM 3 LOGIC CIRCUIT DESIGN PROJECT

Govt. Model Engineering College

Built-In Self-Test (BIST)
4-Bit Adder Implementation

Autonomous hardware verification using LFSR pattern generation, response compaction, and signature analysis.

PROJECT TEAM

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CUT IC

74LS83 4-BIT BINARY ADDER

SYSTEM OVERVIEW

Autonomous Hardware Testing Without External Equipment

The BIST Concept

Eliminates costly external Automatic Test Equipment (ATE) by embedding test vector generation and output checking directly on-chip.

Key Target: VLSI & Memory Testing

Core Mechanism

An 8-bit LFSR generates 127 pseudo-random patterns. Responses are compacted into a 4-bit signature and matched against a golden code.

Golden Signature: 1110 (SIG3-0)

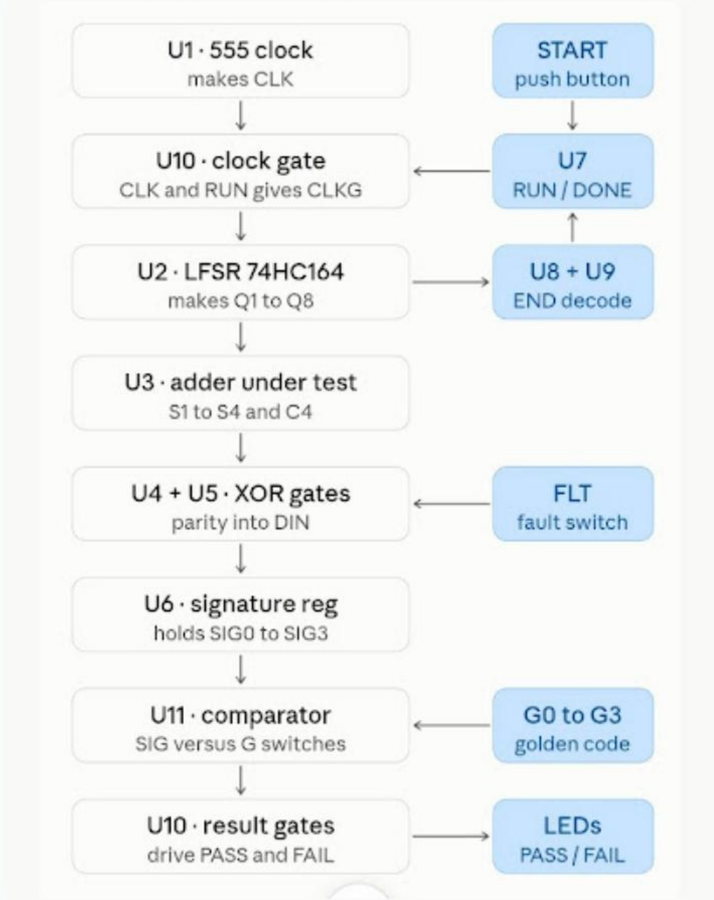
Live Verification

Features an inline fault switch to deliberately invert S1 output, demonstrating immediate red FAIL LED feedback upon error detection.

Fault Coverage: 31/31 Faults Tested

ARCHITECTURE

High-Level System Block Flow



1. Pattern Generation

U1 (555) clocks U2 (74HC164) LFSR to drive A1-A4 and B1-B4 adder inputs.

2. CUT & Response Compaction

U3 (74LS83) adder outputs S1-S4 & C4 are XORed into single parity bit DIN.

3. Signature Analysis

U6 signature register accumulates code over 127 clock ticks.

4. End Control & Comparison

U8+U9 decode 0000011 state to stop clock. U11 compares final signature.

BILL OF MATERIALS

Integrated Circuits Specification (11 ICs)

IC ID	Part Number	Functional Role	Key Connection Detail
U1	NE555	System Clock Generator	Adjustable frequency ~7 Hz to 140 Hz
U2	74HC164	LFSR Pattern Generator	8-bit serial-in shift register (Q1-Q8)
U3	74LS83A	**Circuit Under Test (CUT)**	4-Bit Binary Full Adder
U4, U5	74LS86 (x2)	XOR Feedback & Compaction	LFSR FB, Fault Inject, Parity Tree, SIG FB
U6	74HC164	Signature Register	4-bit response accumulator (SIG0-SIG3)
U7	74LS76A	State Control Flip-Flops	Dual JK flip-flops for RUN and DONE states
U8, U9	74LS27 / 74LS11	End-of-Test Decoder	NOR/AND gates decode pattern 0000011
U10, U11	74LS00 / 74LS85	Comparator & Output Logic	Clock gating, signature compare & PASS/FAIL drive

SUB SYSTEM 1

LFSR Pattern Generation (74HC164)

Feedback Polynomial Logic

Uses primitive polynomial $x^7 + x + 1$. Implemented via XNOR between Q1 and Q7 stage outputs.

```
FB = Q1 XNOR Q7 = Q1 XOR (NOT Q7)
```

- XNOR feedback permits valid operation from the all-zero start state.
- Cycles through 127 unique pseudo-random patterns automatically.

Adder Vector Mapping

- **Input A (4-bit):** Driven by Q1, Q2, Q3, Q4
- **Input B (4-bit):** Driven by Q5, Q6, Q7, Q8
- **Carry In (C0):** Tied to GND (0V)

Initial State Sequence

```
Tick 0: 0000000 | Tick 1: 1000000
Tick 2: 0100000 | Tick 3: 1010000
Tick 4: 0101000 | Tick 5: 1010100
```

SUB SYSTEM 2

Response Compaction & Fault Injection

5-Bit Space Parity XOR Tree

Squeezes 5 parallel outputs (S1-S4 + C4) into 1 bit stream per clock tick.

```
S1F = S1 XOR FLT (Fault Injection Gate)
```

```
X1 = S1F XOR S2 | X2 = S3 XOR S4
```

```
X3 = X1 XOR X2 | P = X3 XOR C4
```

Compacted Output P = S1F ⊕ S2 ⊕ S3 ⊕ S4 ⊕ C4

FLT Fault Toggle

SPDT switch controls U4 Gate 2B. Inverts S1 output to simulate physical hardware failure.

- **FLT = GND:** Normal operation (Pass)
- **FLT = +5V:** Inverted S1 (Triggers Fail)

SUB SYSTEM 3

Signature Register & Time Compaction

Serial-In Feedback Register

U6 (74HC164) acts as a 4-bit signature analyzer utilizing polynomial $x^4 + x^3 + 1$.

```
F03 = SIG0 XOR SIG3
```

```
DIN = P XOR F03 (Shifted into Pin 1)
```

```
Outputs: SIG0 (QA), SIG1 (QB), SIG2 (QC), SIG3 (QD)
```

EXPECTED GOLDEN SIGNATURE

1110

SIG3 = 1 | SIG2 = 1 | SIG1 = 1 | SIG0 = 0

SUB SYSTEM 4

Control Logic & Automatic Termination

STEP 01

Initialization

Pressing /START clears shift registers and presets RUN flip-flop (U7 Pin 15 = HIGH).

STEP 02

Clock Gating

U10 NAND gates 555 CLK with RUN signal to generate gated clock (CLKG).

STEP 03

END Decoding

U8 NOR + U9 AND detect LFSR state `0000011` (Pattern 126).

STEP 04

Auto Stop

DONE flip-flop sets, clearing RUN. Clock stops automatically at 127 ticks.

VERIFICATION

Comparison & Pass/Fail Display

74LS85 4-Bit Magnitude Comparator

Compares signature register bits (A = SIG0-SIG3) against golden reference switches (B = G0-G3).

```
PASS_raw = (A == B) [Pin 6 Output]
```

Output gated by FINISH signal (FINISH = DONE AND nRUN) so LEDs light only when test completes.

GREEN PASS LED
`Signature == 1110 (Fault-Free)`

RED FAIL LED
`Signature ≠ 1110 (Fault Injected)`

SIMULATION DATA

Experimental Fault Coverage (31 Faults Tested)

Test Condition / Fault Type	Tied Node / Trigger	Final Signature	Visual Output
Fault-Free Adder	FLT Switch = GND	**1110**	**GREEN PASS**
Fault Switch Active	FLT Switch = +5V (S1 Inverted)	**1101**	**RED FAIL**
S1 Stuck-at-0 / Stuck-at-1	U4 Pin 4 tied to GND / +5V	1001 / 1010	**RED FAIL**
S2 Stuck-at-0 / Stuck-at-1	U4 Pin 10 tied to GND / +5V	1000 / 1011	**RED FAIL**
S3 Stuck-at-0 / Stuck-at-1	U4 Pin 12 tied to GND / +5V	0000 / 0011	**RED FAIL**
S4 Stuck-at-0 / Stuck-at-1	U4 Pin 13 tied to GND / +5V	0100 / 0111	**RED FAIL**
C4 Stuck-at-0 / Stuck-at-1	U5 Pin 5 tied to GND / +5V	0101 / 0110	**RED FAIL**

ANALYSIS

Engineering Trade-offs & Future Scope

BIST Limitations

- **Aliasing Probability:** 4-bit signature register has ~1/16 (6.25%) chance of missing a fault.
- **Pseudo-Random Coverage:** 127 patterns cover ~50% of total 256 input states (non-exhaustive).

$$\text{Aliasing } P = 1 / (2^n) = 1 / (2^4) = 0.0625$$

Proposed Upgrades

- **8-Bit Signature Analyzer:** Expand signature to 8 bits using dual 74LS85s, dropping aliasing to 1/256 (0.39%).
- **MISR Architecture:** Implement Multi-Input Signature Register for parallel response capture.

$$\text{Target Aliasing } P = 1 / (2^8) = 0.0039$$

PROJECT DEMO COMPLETE

Thank You! Questions?

Demonstrating Autonomous BIST Principles with Discrete 74-Series Logic ICs & NE555 Timer.

✓ 127 Test Patterns ✓ Golden Sig: 1110 ✓ Live Fault Switch